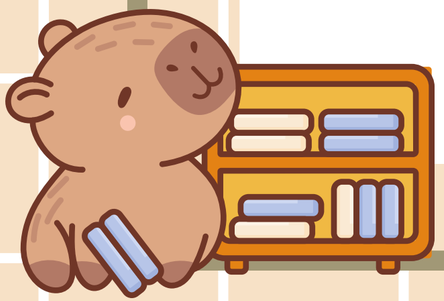
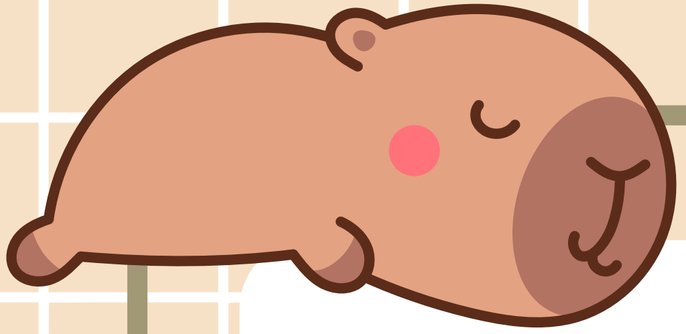
ZOOTOPIA MANAGEMENT SYSTEM

Objectives

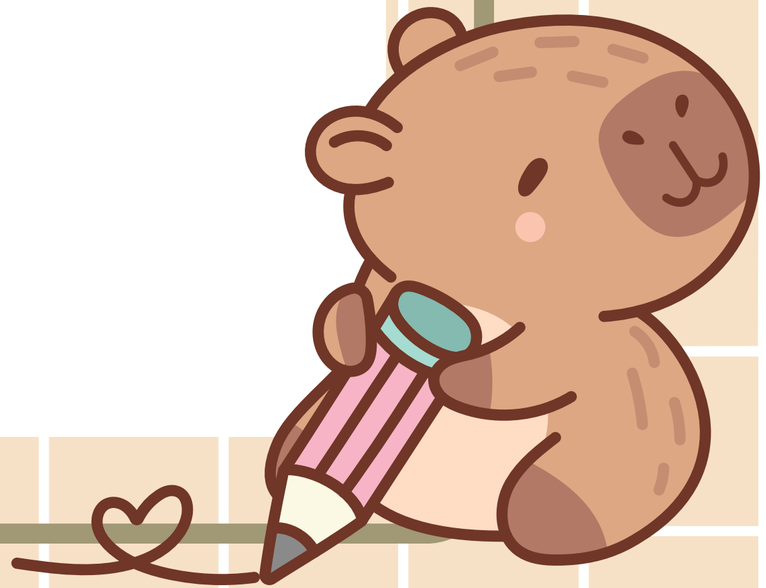
- To develop a zoo management system using Java Swing
- To apply Object-Oriented Programming (OOP) concepts
- To manage animal data efficiently (add, remove, view)
- To create a user-friendly graphical interface

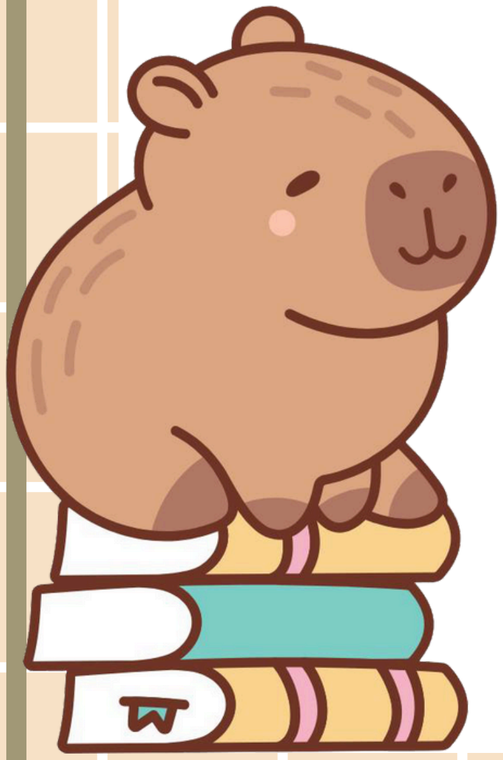




System Features

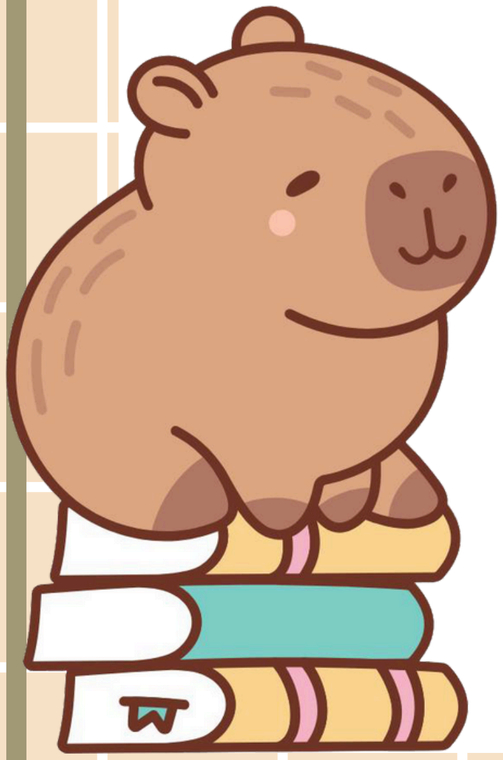
- Manage animal species and store for future use
- Add animal information
- Remove animal information
- Display animal details
- Import animal data
- Interactive dialog-based operations





GUI Components Used

- JFrame
- JPanel
- JButton
- JLabel
- JTextField
- JTextArea
- JComboBox
- JDialog
- JTree
- JScrollPane
- TitledBorder
- ImageIcon



Event Listener Used

- ActionListener (For Button)
- MouseListener (For Button)
- FocusListener (For TextField and TextArea)
- TreeSelectionListener (For Tree)

Dialog & User Interaction

- AddDialog
- RemoveDialog
- ShowDialog
- ImportDialog
- TicketDialog
- UpdateDialog

Main Frame

BorderLayout.NORTH

Image Icon

Image Icon

Title Panel

BorderLayout.WEST

Welcome Panel

Panel with tree and list

JButton to open add species
JDialog to add new species

JButton to open update species
JDialog to update species

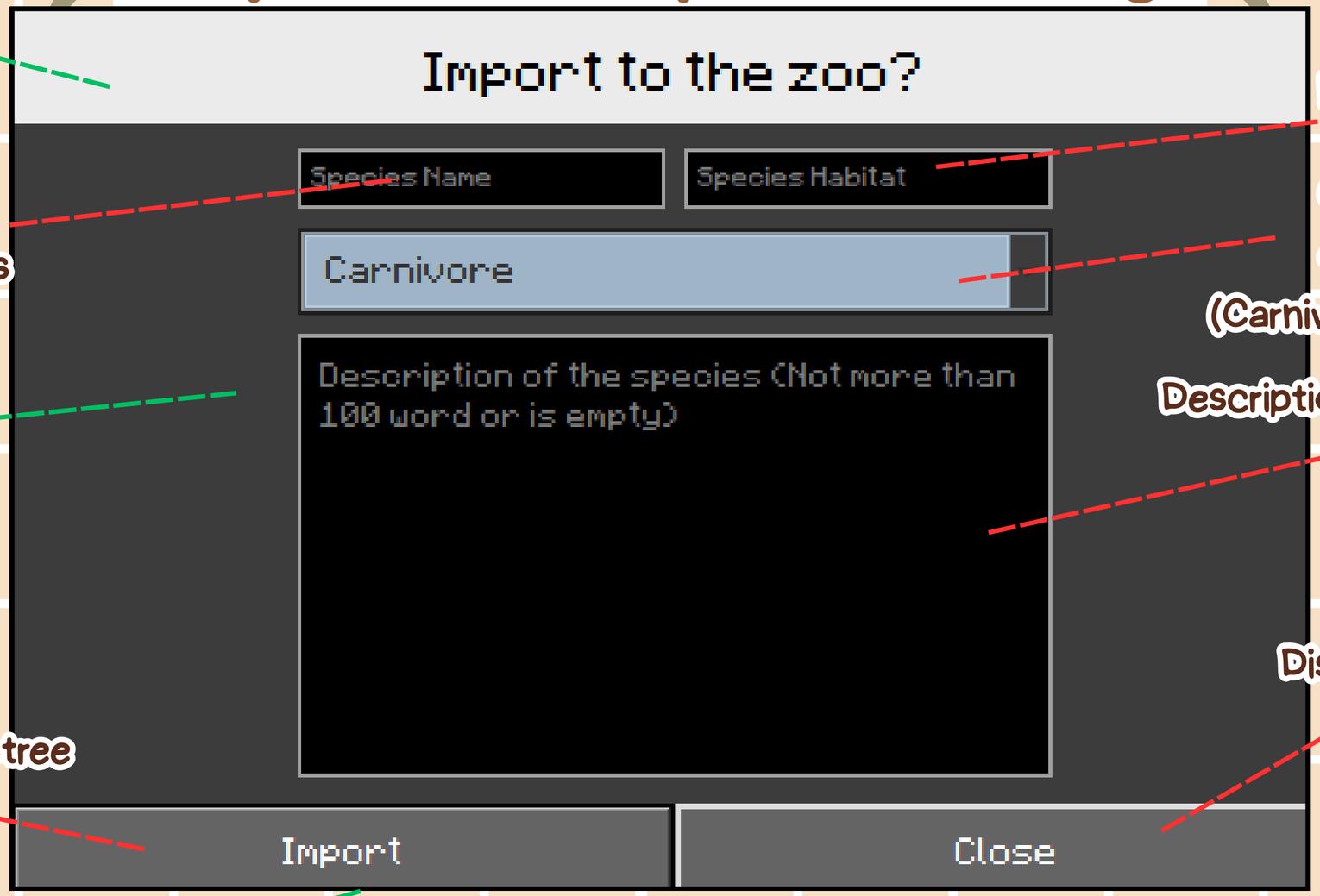
JButton to open remove dialog to remove animal

JButton to open ticket dialog to calculate ticket price

JButton to open add dialog and add new animal

BorderLayout.SOUTH, GridLayout(3, 1)

Import new Species Dialog



BorderLayout.NORTH

TextField for the habitat of the species

TextField for the name of the species

Combobox for the diet of the species (Carnivore, Herbivore, Omnivore)

GridBagLayout and GridBagConstraint

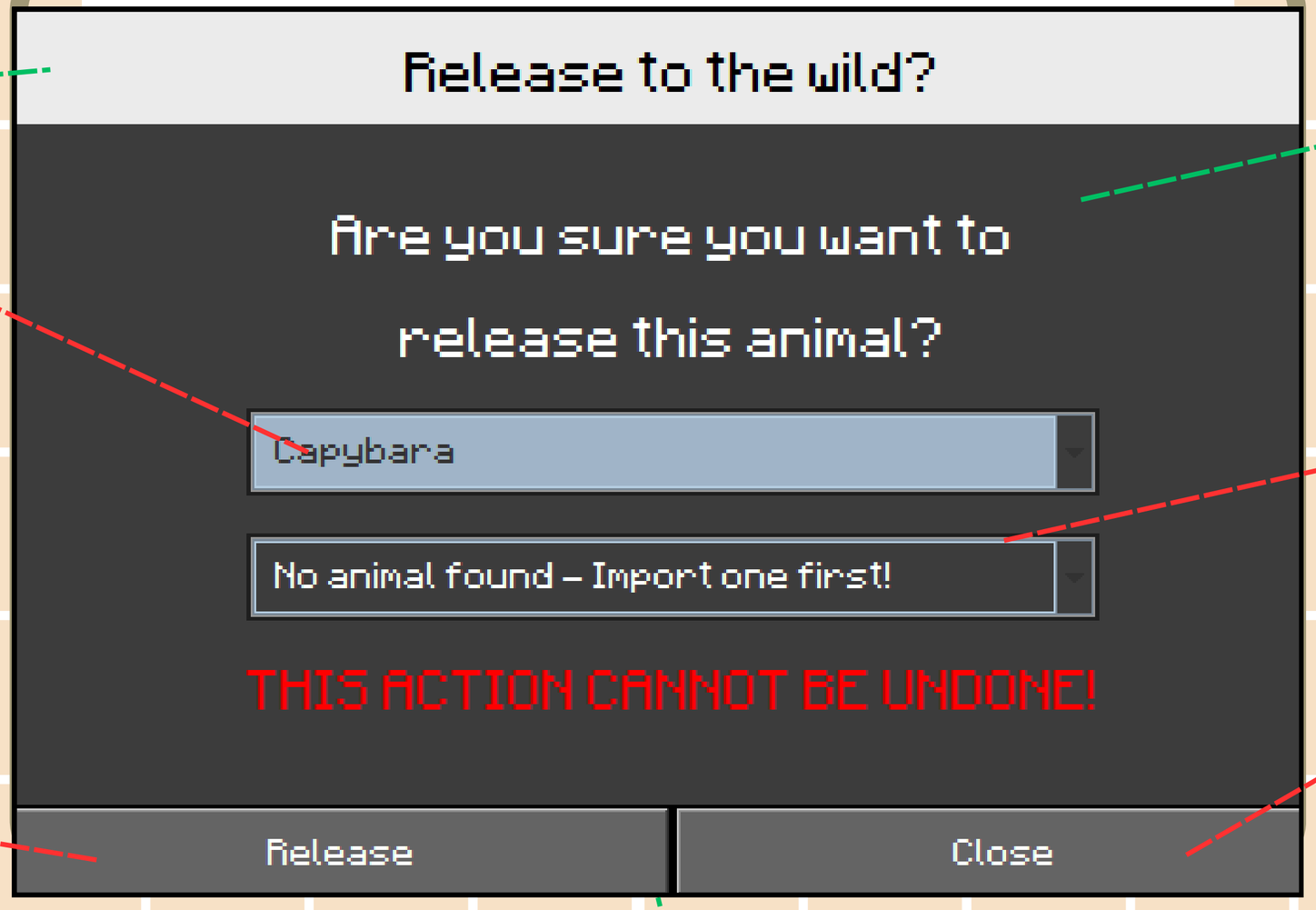
Description area for the species

Import button to save the species into file and update the tree

Dispose the dialog

BorderLayout.SOUTH, GridLayout(1, 2)

Release animal dialog



BorderLayout.NORTH

GridBagLayout
and GridBagConstraint

Combobox for the
available species

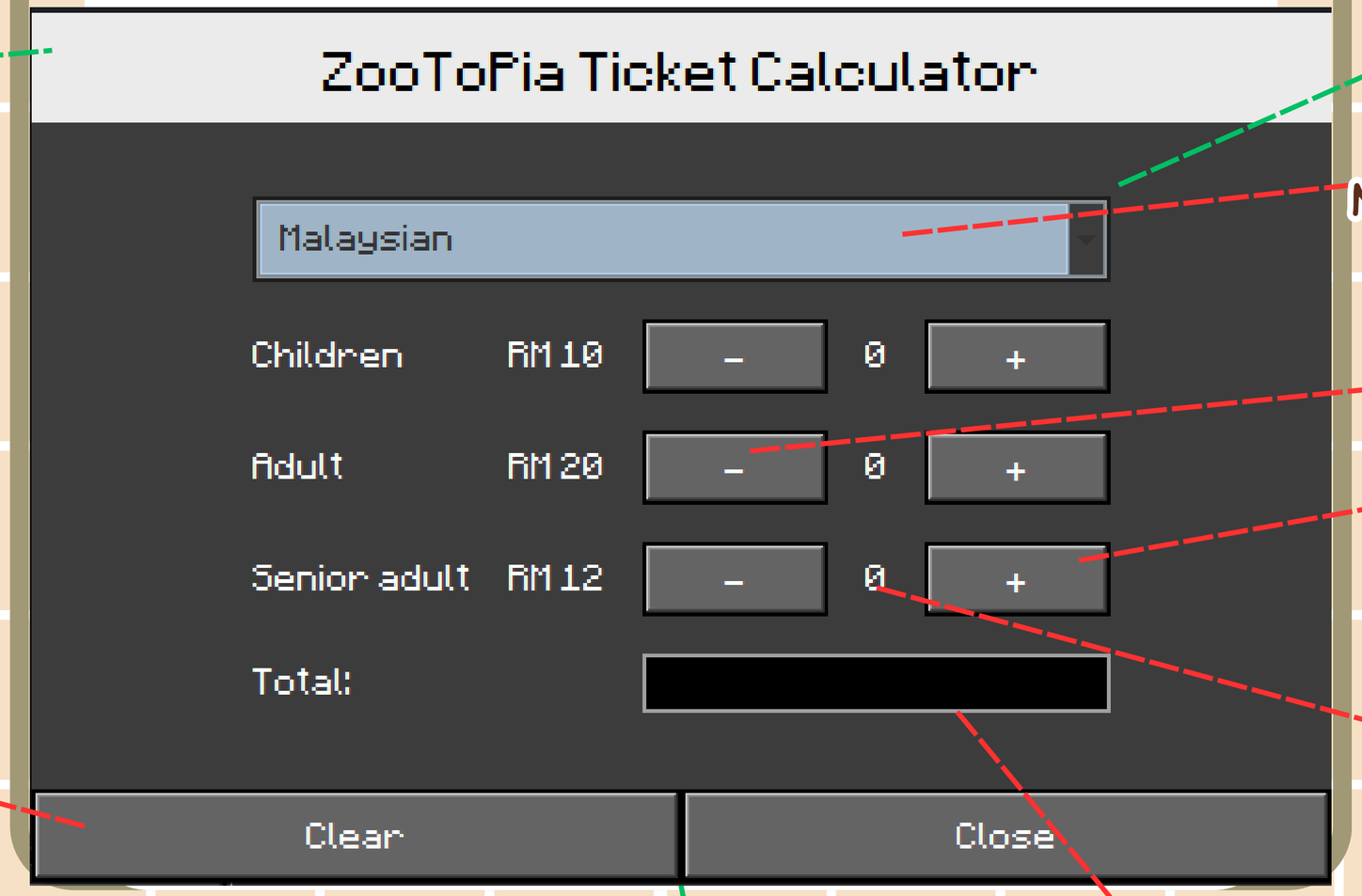
Available animal
for the species

Release button to
remove the animal
from the file
and update the tree

Dispose the dialog

BorderLayout.SOUTH, GridLayout(1, 2)

Zoo Ticket Calculator Dialog



BorderLayout.NORTH

GridBagLayout and GridBagConstraint

ComboBox to choose Malaysian OR Non-Malaysian

Minus button to minus 1

Add button to add 1

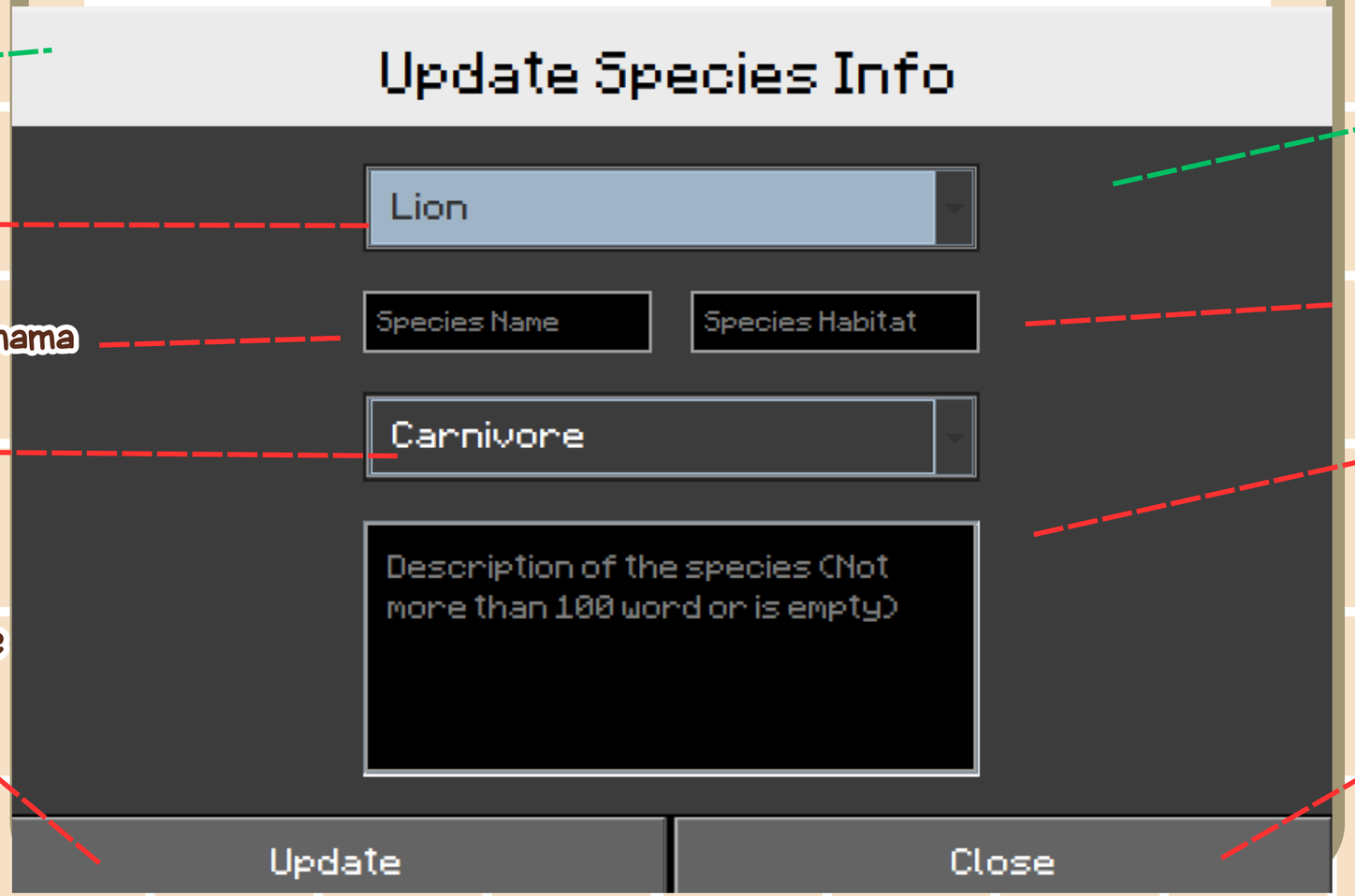
Clear button to clear all the number and calculations

Quantity label that changes follow the user action

Real time total

BorderLayout.SOUTH, GridLayout(1, 2)

Update species dialog



BorderLayout.NORTH

GridBagLayout and GridBagConstraint

Combobox for the available species

Textfield for the species nama

Textfield for the species habitat

Combobox for the available diet

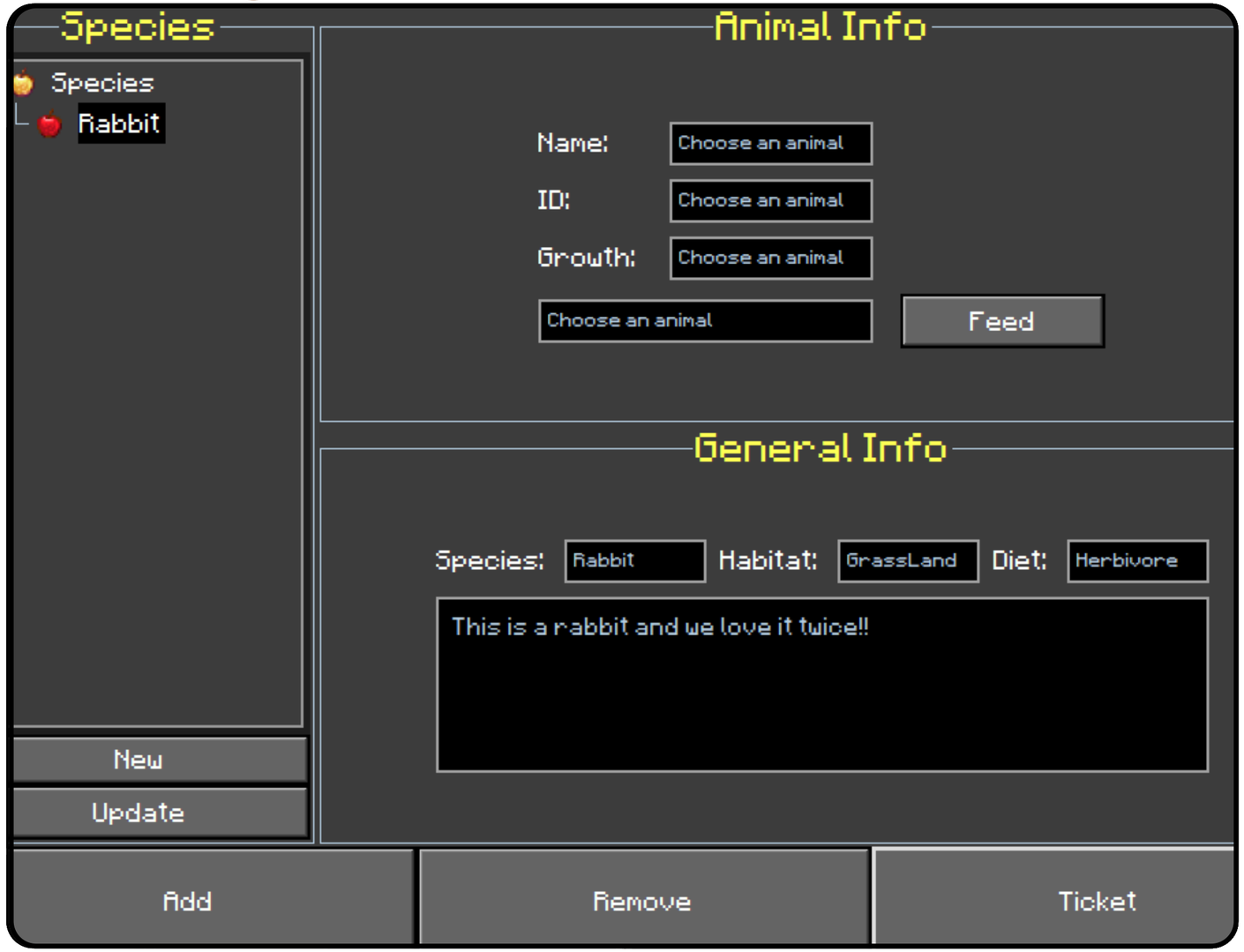
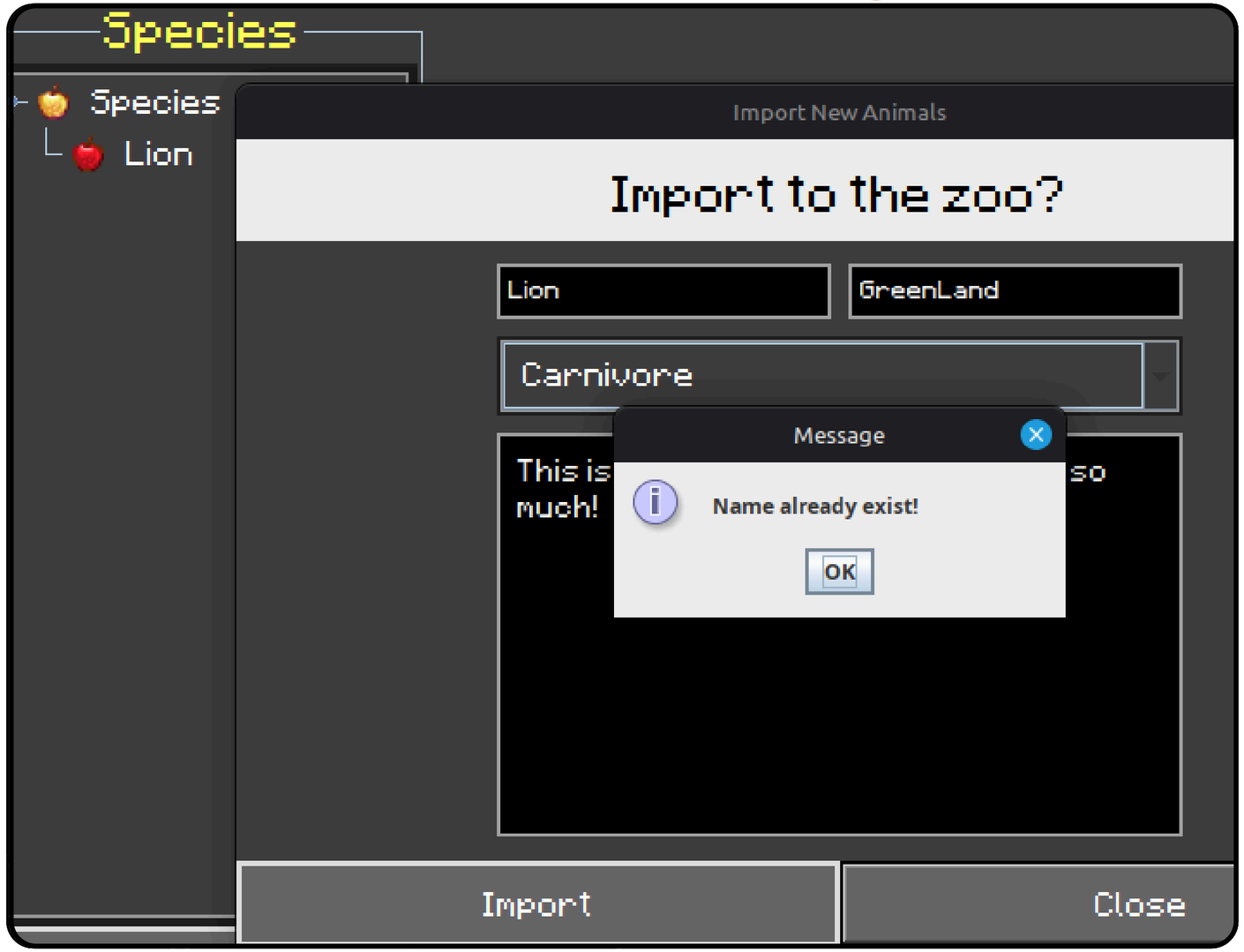
Animal description

Update button to update species info

Dispose the dialog

BorderLayout.SOUTH, GridLayout(1, 2)

System Flow Overview



Error Handling